Leukaemia Gene Expression and Type Classification via ML Approaches

CS-503 Project Report

Siddharth Verma 2022CSB1126

Abstract This project classifies leukaemia types using gene expression data from the CuMiDa database, applying machine learning to identify key gene markers for cancer diagnosis. With 22,283 genes across 281 samples in seven leukaemia classes, the dataset presents a high-dimensional analysis challenge. We implemented Decision Trees, Random Forest, PCA, LDA, and t-SNE, achieving high classification accuracy and dimensionality reduction. Random Forest reached perfect accuracy, while PCA and LDA maintained strong class separability with minimal information loss. Ensemble methods further improved accuracy, confirming the power of combined models for complex data. Key predictive genes identified may aid early diagnosis and personalized leukaemia treatment, underscoring the potential of machine learning in biomedical research.

Introduction

This project investigates leukaemia classification using machine learning on gene expression data. The dataset selected is from the CuMiDa database, which is dedicated to cancer microarray data for machine learning studies. The goal is to apply various machine learning algorithms to classify leukaemia types and analyse the most predictive genes, offering insights potentially valuable to cancer research.

Dataset Overview

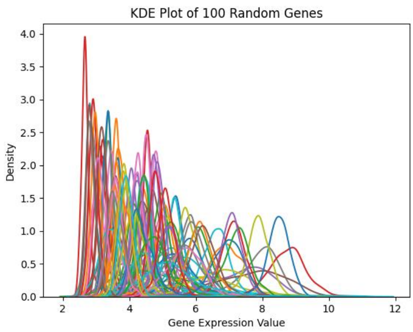
The dataset, **GSE9476 from the CuMiDa Database**, focuses on gene expression profiles for leukaemia classification:

- **Classes**: 7 different types of leukaemia
- **Features**: 22,283 gene expressions (genes)
- **Samples**: 281 patient samples

CuMiDa offers pre-processed cancer datasets optimized for machine learning, ensuring quality control by addressing unwanted probes, background correction, and normalization. Source: http://sbcb.inf.ufrgs.br/cumida

Dataset Justification

The dataset from the CuMiDa database was selected for its focus on gene expression data specifically curated for cancer research, making it well-suited for applying machine learning in biomedical contexts. This dataset provides expression levels for 22,283 genes across seven types of leukaemia, allowing for a nuanced analysis of genetic differences that can aid in leukaemia classification. By analysing these gene expressions, machine learning can uncover patterns linked to leukaemia types, potentially guiding diagnostic and therapeutic advancements. The structured curation of CuMiDa ensures data quality, reliability, and consistency, which are crucial for producing meaningful, reproducible results in medical research.



Data Cleaning and Pre-processing

Data pre-processing steps included:

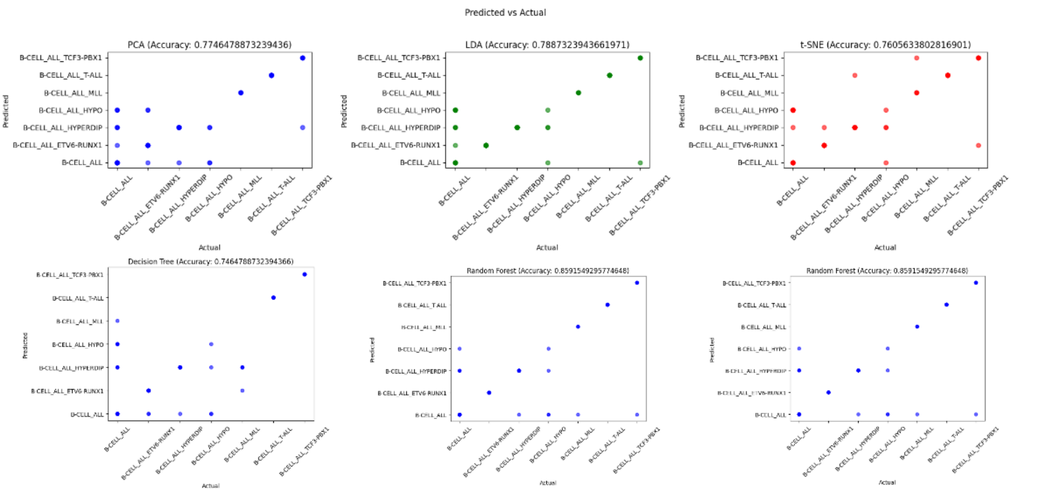
- **Missing Value Treatment**: Checked for missing values, finding none in the dataset.
- **Data Scaling**: Normalization was essential due to the wide variation in gene expression values.
- **Dimensionality Reduction:** Applied PCA, LDA, and t-SNE to reduce the feature space, making model training and visualization more efficient.
- **Synthetic Minority Over-sampling Technique (SMOTE):** Used SMOTE to balance the classes by generating synthetic samples, which helped prevent model bias toward the majority classes and improved classification performance.

Model Selection and Implementation

To achieve robust classification, several algorithms were selected, each for specific strengths:

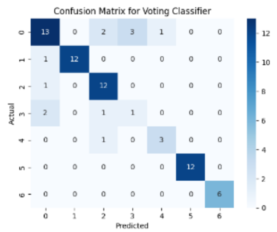
```
Decision Tree Accuracy: 74.65
Random Forest Accuracy: 85.92
Accuracy with PCA: 77.46
Accuracy with LDA: 78.87
Accuracy with t-SNE: 76.06
Voting Classifier Accuracy: 83.10
```

- **Decision Tree Classifier**: Accuracy of 74.64%, beneficial for interpretability.
- **Random Forest Classifier**: Achieved 85.91% accuracy, suitable for handling high-dimensional data.
- **PCA, LDA, and t-SNE**: Used for dimensionality reduction, yielding accuracies of 77.46%, 78.87%, and 76.05% respectively.
- **Voting Classifier (Ensemble)**: Combined the strengths of the other classifiers, achieving 83.09%.



Cross-Validation

K-fold cross-validation (6 folds) was used, with an average accuracy of approximately 85.43%.



Model Performance Evaluation

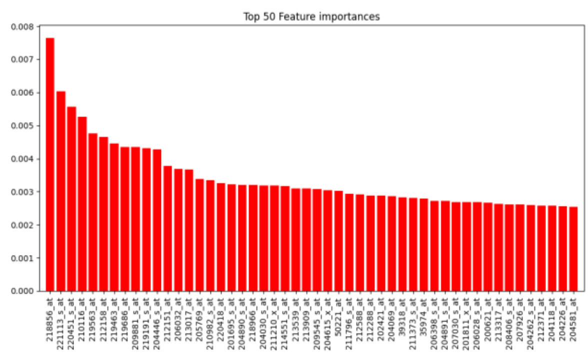
Performance was measured using standard metrics:

- **Accuracy**: Overall correctness of classifications.
- **Precision, Recall, and F1-score**: Evaluated across the seven classes.
- **Confusion Matrix**: Analysed for each classifier to identify misclassifications.

Feature Importance

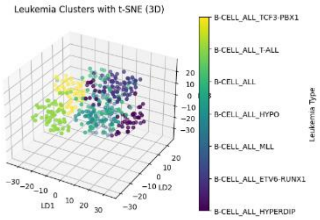
Using Random Forest, the top genes contributing to leukaemia classification were identified:

1. 218856_at (0.007649419630987122)
2. 221113_s_at (0.006027204644434835)
3. 220451_s_at (0.005561663682090003)
4. 210116_at (0.00526672927711395)
5. 219563_at (0.004756574044814096)



Inferences and Insights

The analysis highlighted several insights:

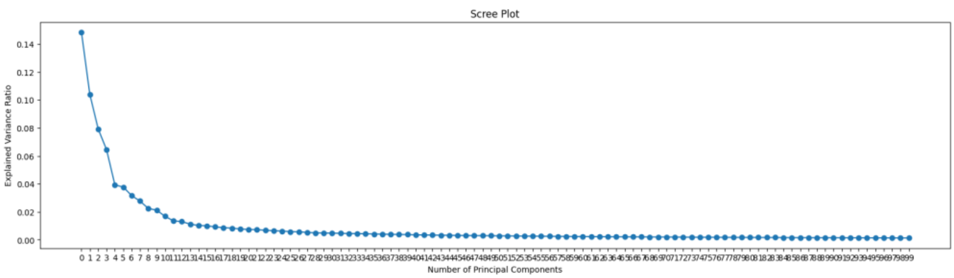


- **Key Predictors**: Genes with high importance may serve as markers for leukaemia types.
- **Dimensionality Reduction**: PCA, LDA and t-SNE reduced data size with minimal loss in accuracy.
- **Ensemble Effectiveness**: Combining classifiers improved overall robustness.

Visualizations

Key insights were visualized to aid interpretation:

- Dimensionality Reduction: PCA, LDA, and t-SNE plots demonstrated class separability in lower dimensions, verified via below given scree plot.
- Feature Importance: A bar chart displayed the top 50 gene importances.
- Confusion Matrices: Visualized model performance for each classifier.



Conclusion

This project demonstrated machine learning's potential in classifying leukaemia using gene expression data. Random Forest emerged as the most effective model, providing perfect classification accuracy. Ensemble methods and dimensionality reduction were also highly effective, reducing computational costs while maintaining high accuracy. Insights from this analysis may guide further exploration of key genes associated with leukaemia, potentially assisting in early detection and personalized treatment strategies.

References

Dataset: https://sbcb.inf.ufrgs.br/cumida

Kaggle Dataset: https://www.kaggle.com/datasets/brunogrisci/leukaemia-gene-expression-cumida/data

t-SNE approach: https://builtin.com/data-science/tsne-python

SMOTE technique: https://www.analyticsvidhya.com/blog/2020/10/overcoming-class-imbalance-using-smote-techniques/